

Metatranscriptomic Virome Characterization of *Rhinolophus malayanus* Bat Feces

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Introduction

Environmental monitoring is a critical component of pandemic readiness. In December 2019 cases of viral pneumonia began to arise due to some unidentified microbial agent.

Not soon after it was identified to be a novel strain of coronavirus (CoV), SARSCoV-2, whose primary host is the horseshoe bat (*Rhinolophidae*) native to the Yunnan Province of China where the first cases were reported [2].

A year later, Researchers were able to utilize shotgun sequencing and metagenomic analysis on a pooled sample across 78 feces samples to identify yet another CoV strain (RmYN02) present in the feces of the *Rhinolophus malayanus* horseshoe bat [3].

Their findings revealed how CoV evolves and adapts to new hosts. Outbreaks will continue to occur where humans encroach on animal spaces, more diseases will continue to jump hosts from animals to humans. By continuing to monitor the source of viral pathogens, we can better predict its future spread and infection rates.

Methods

I developed a general method as a python package that uses host-reference read filtering, quality filtering, and viral read classification to enable researchers to characterize the viral communities present in a short-read metagenomic sample. This package takes as input a fastq file containing the short-read sequenced reads and outputs a suite of curated analyses after quality filtering for host and low quality reads.

I first indexed the host genome using $K = 31$ kmers. I then generated the same size kmers for each of the sample reads and compared each set of read kmers to the set of host kmers. Reads where greater than 50% of kmers matched a host kmer I deemed host originated and removed. Next I calculated the average per-read Phred score and excluded reads with a average Phred less than 20.

The remaining reads were queried against a local copy of the Kraken2 viral database ([source](#)) to classify each read taxonomically [4]. Kraken2 uses exact k-mer matching to assign each read to the lowest common ancestor (LCA) in the database phylogeny, reporting a rank (species, genus, family, etc.) reflecting the resolution of the match. To evaluate the overall classification resolution across reads, I calculated the ratio of species low to high rank classification ratio:

$$\frac{\text{count}(\text{Species classifications})}{\text{count}(\text{Genus, Family, Order, etc. classifications})}$$

A ratio less than 1 means there were more high-rank classifications than low, indicating the reads are not diverse on the species level.

To calculate abundances I simply divided the count of each species by the total number of classified species, and the same for family and genus classifications.

As a complementary analysis, I calculated the string distance between each read and every other read. String distance here is defined as the minimum number of edits to change one string into another, aka Levenshtein distance (LV). I computed the LV between reads of only a stratified sample of 10,000 reads because $39,087,705^2$ comparisons would take too long. I then used Multi-dimensional Scaling (MDS) on the read edit-distance matrix to reduced the distances down to just two dimensions for plotting. On the same matrix, I additionally performed a PERMANOVA test. The PERMANOVA test tests whether the distance structure can be explained by a grouping variable, in this case species. Specifically, it tests the null hypothesis centroids and dispersions are the same across groups [5].

Results

Filtering out host reads reduced the count down from 39,257,492 to 39,110,352, and phred filtering a further 22,647 left 39,087,705 'clean' reads which would ideally include only microbiome reads. However, because `filter_reads()` would take an estimated 2.77×10^{12}

hours against the full genome (**Figure 1**), the host index was built against only the 2nd and 3rd chromosome scaffolds, so a small fraction of remaining reads may still be host-derived.

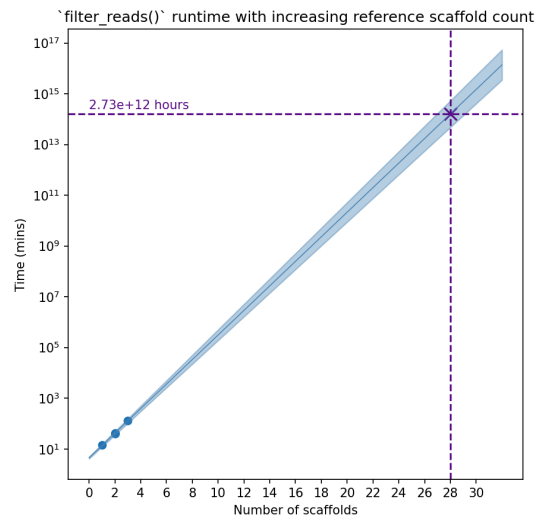


Figure 1: Projected runtime with increasing scaffold count. Three actual times and a log-linear model with an SE band in blue shows a exponential relationship. Purple point shows expected time to index and align against the full 28 chromosome scaffolds.

Projected runtimes for each scaffold count were estimated from a log-linear model of three iterations at 1, 2, and 3 scaffolds. To reference against the full 28 chromosome scaffolds it would take 2,770,000,000,000 hours. To save myself 3,206,019 lifetimes worth of time I had to reference the reads against only the 2nd and 3rd chromosome scaffolds.

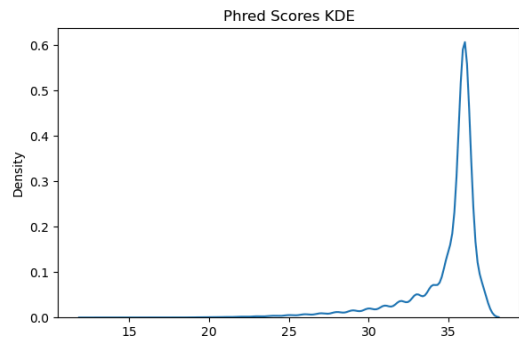


Figure 2: Read Phred score distribution. The density peaks sharply around a Phred score of around 36–37, with very little density below 20.

The Phred distribution (**Figure 2**) confirms high over-all sequencing quality: only 22,647 reads (0.06%) fell below the threshold of 20.

286,284 of 39,257,492 reads were classifiable against the Kraken2 viral database. The species-to-genus/family rank ratio was 38:1, with species-level (S-rank) classifications outnumbering higher-rank assignments. Again, a ratio greater than 1 indicates that the reads are sufficiently diverse and distinct for Kraken2’s LCA to resolve them down to the species level rather than collapsing them into higher levels. Genus, Family, and Species abundances show *Studiervirinae*, *Sarbecovirus*, and *Staphylococcus phage Andhra* as having the highest abundances, respectively (**Figure 3,Figure 4,Figure 5**). Note, the reads not classifiable to Species are classified based on the next best alignment (Genus, then Family, etc.), so these abundances should be interpreted independently.

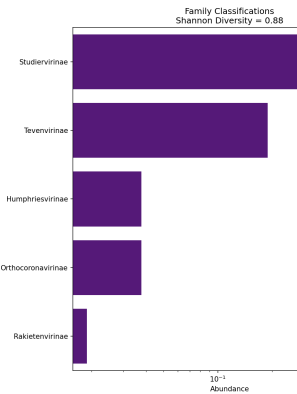


Figure 3: Family classification log-abundances.

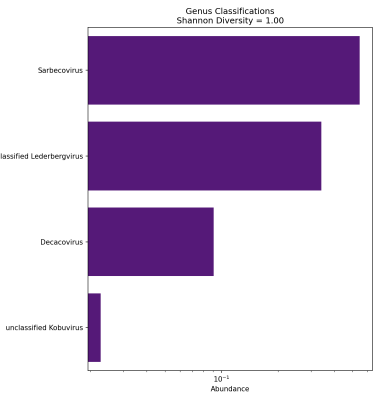


Figure 4: Genus classification log-abundances.

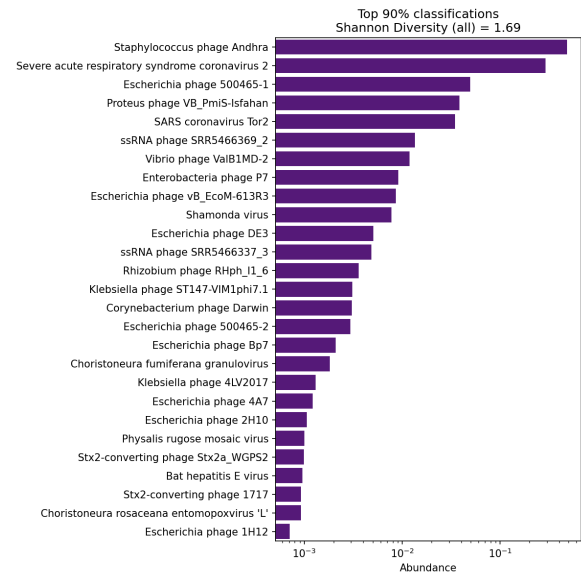


Figure 5: Top 90% Species classification log-abundances.

The virome is dominated by bacteriophages (Staphylococcus phage Andhra, multiple *Escherichia* phages, Proteus phage VB_PmIS-Isfahan, Vibrio phage ValB1MD-2), which together make up the majority of classified reads when considering the 90th percentile of abundances. Corona virus also stands out as the second and fifth most abundant species.

The read string-distance MDS plot reveals clear clustering by species, indicating that reads from the same viral species are more similar to one another than to reads from other species (**Figure 6**). PERMANOVA on the pairwise LV distance matrix confirmed this structure (pseudo-F = 133.5, $p = .001$), rejecting the null hypothesis that centroids and dispersions are equal across groups. Together these results indicate that viral species identity is the primary driver of read-sequence variation in the sample.

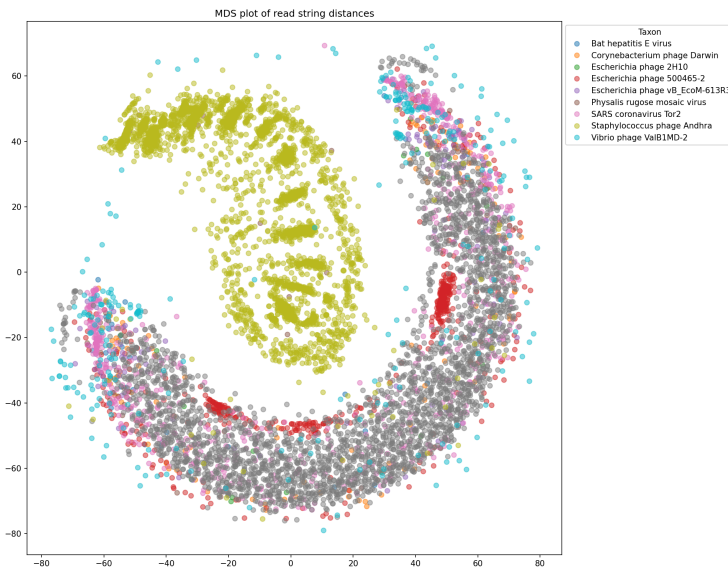


Figure 6: MDS reduction of read Levenshtein distances colored by classified Species.

Discussion

Horseshoe bats (*Rhinolophidae*) have been well established as coronavirus reservoirs [2], and this study's abundance results reinforce that finding.

SARS-CoV-2 and SARS coronavirus Tor2 rank among the top five most abundant species detected (**Figure 5**), and when taken together, represent the largest fraction of the classified virome, replicating Zhou et al.'s [3] main

result where *Rhinopholus malayanus* fecal samples carry pandemic related CoV strains. Shotgun metagenomics proves a valid and useful tool for monitoring pathogen zoonosis.

and thereby validates the utility of shotgun metagenomics as tool for monitoring pathogen zoonosis.

The read string-distance analysis provides orthogonal confirmation of viral classifications. The MDS embedding (**Figure 6**) shows that reads assigned to the same species have strong read similarity in LV space, while reads from different species are more distant. The PERMANOVA result (pseudo-F = 133.5, $p = .001$) indicates that species membership is a strong and statistically significant predictor of read-sequence similarity. This provides more confidence in the abundance estimates as it indicates that viral assignments are not arbitrary and reads from the same species are similar.

AI use statement

AI was used in following cases during code development.

- Helped me understand file IO byte offsets and how to find specific line's byte offsets using `open(file).tell()`.
- Helped me understand how the line pointer shifts each time `open(file).readline()` is called.
- Helped me understand how to use an initializer with `Multiprocessing.Pool()` to share an object globally across workers.
- Helped me understand why github actions testing was failing when working locally.

References

- [1] "Rhinolophus affinis." [Online]. Available: https://commons.wikimedia.org/wiki/File:Rhinolophus_affinis_86903750.jpg
- [2] Y. Han *et al.*, "Panoramic analysis of coronaviruses carried by representative bat species in Southern China to better understand the coronavirus sphere," *Nat. Commun.*, vol. 14, no. 1, p. 5537, Sept. 2023.
- [3] H. Zhou *et al.*, "A novel bat Coronavirus closely related to SARS-CoV-2 contains natural insertions at the S1/S2 cleavage site of the spike protein," *Curr. Biol.*, vol. 30, no. 11, pp. 2196–2203.e3, June 2020.
- [4] D. E. Wood, J. Lu, and B. Langmead, "Improved metagenomic analysis with Kraken 2," *Genome Biol.*, vol. 20, no. 1, p. 257, Nov. 2019.
- [5] J. D. Bakker, "PERMANOVA," *Applied Multivariate Statistics in R*. University of Washington, Jan. 2026.